

Performance Testing

Date: 30 September 2026
Team ID: SWTID-2026-3656
Project Name: EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks: 5 Marks

Testing Overview

Field	Actual Details
Tool	Browser, FastAPI TestClient, curl, Uvicorn logs
Type	Functional smoke testing and startup verification
Target	/, /explain, /ask, /quiz, /summarize, /learning-path
Environment	Local sandbox and Windows-compatible commands
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Test Scenarios

No.	Scenario	Expected Outcome	Observed
1	GET /	HTML homepage, status 200	Pass
2	POST /explain with gravity	LaMini explanation, status 200	Pass
3	POST /ask with placeholder key	Safe configuration error	Pass
4	POST /quiz	Safe provider response or validated quiz	Pass
5	POST /summarize	Safe provider response or summary	Pass
6	POST /learning-path	Safe provider response or path	Pass
7	Empty input	Useful validation response	Pass

Results

Metric	Target	Result	Status
Startup	Uvicorn starts	Application startup complete	Pass
Homepage	HTTP 200	HTTP 200	Pass
Static CSS	HTTP 200	HTTP 200	Pass
Local explanation	Valid response	Valid LaMini response	Pass
Gemini configuration	No secret leakage	Safe 503 message when absent	Pass
Error handling	Useful errors	JSON error responses	Pass

The first local LaMini request downloads model weights and may require several GB of disk space. Cloud Gemini features depend on a valid key, model availability, and network access.